Sunday 7th July, 6.00pm

Many Worlds Theory **(11'10")**

Gordon Monro

The *many-worlds theory* of H. Everett is an interpretation of quantum physics. According to this theory, every time a random event could have happened, the universe splits into two: in one copy of the universe the random event happened; in the other it did not. Thus there are an alternative number of "alternative universes", with new ones splitting off all the time. Everett advanced this theory as a replacement for the probabilistic interpretation of quantum mechanics, which is what Einstein was objecting to when he said "God does not play dice". In this piece the alternative universes are represented by two slowly diverging pitches and by randomly arranged snippets from computer games, the alternative universes most accessible to us.

Many Worlds Theory has as framework part of a work for solo harpsichord by J.S. Bach (the *Adagio* from BWV 974). This work is itself a transcription from the different world of an oboe concerto by the Venetian composer Alessandro Marcello (though it was earlier thought to be a transcription of a Vivaldi violin concerto). The original oboe melody is represented in the tape part in greatly slowed-down form (like the Chinese melodies in Japanese T_gaku court music); much of the harpsichord part consists of computer-generated elaborations of parts of Bach's elaboration of the oboe melody.

Many Worlds Theory is dedicated to the harpsichordist extraordinaire **Winsome Evans**, who offered much help and encouragement during the composition of the piece, and who premiered the work on October 13, 2001 at the Old Darlington School, University of Sydney.

Gordon Monro has been composing electronic and computer music for about 10 years; many of his pieces have a connection with mathematics. More recently he has been working on a series of pieces for instrument and tape. Gordon completed a Graduate Diploma in Composition at the University of Sydney in 1997 and was granted associate representation as a composer with the Australian Music Centre in 1998.

Emanations (7'00")

Bruce Schneider

This piece was constructed in real time using MIDI note and controller data on the EMU Proteus 2000 synth module. Seven notes (c1, g1, c2, g2, c3, c4, c5) were allocated to seven different patches (sounds) on the Proteus and sustain on commands inserted so these sounds would effectively not stop until faded out. Up to 16 parameters in each sound may be altered in real time by controller data. This was sent to the Proteus using an analogue to MIDI converter with 16 separate control sliders. Parameters that could be controlled included tone, presence, sound shape, imaging, attack ,decay, movement between layers, oscillator rates, volume and sends to internal effects processors. The MIDI data was recorded in real time using "Cakewalk" and the performance repeated until the desired outcome was achieved. The piece you are hearing was the seventh attempt.

Bruce Schneider is a composer/audio engineer/music arranger living in Maitland (30 minutes west of Newcastle NSW) and has been involved with electronic music since the mid 1980's. He was also a part of the composers collective Trojan Theatre until the late 1990's along with fellow members Greg Jenkins, Eric Wilding, Ben Sutures and Noel Burgess.

Second Thoughts *(8'18")*

Dennis Miller

Second Thoughts was composed in 2000 and is intended for performance on videotape. The work is in three sections, the first two of which dominate the form. The opening section explores the inside of a virtual object and depicts many of the surfaces and textures found therein. The second section moves into 3D space and presents different perspectives of the initial object, as well as adding a number of new elements that derive their form from the elements in the opening. The short, third section is a recap of the first and adds several minor variations to it. The music, also composed by this author, contributes an emotive element as well as an added layer of continuity to the piece.

Like previous works by this author, all visual elements were scripted using the POVray scene description language; no special effects or plug-ins of any type were used. The musical material derives from both synthetic and acoustic sound sources.

Dennis Miller received his Doctorate in Music Composition from Columbia University in 1981. Since that time, he has been on the Music faculty of Northeastern University in Boston where he heads the music technology program and serves on the Multimedia Studies Steering Committee. He is currently Associate Professor. Miller was the founder and served as director of the League-ISCM in Boston from 1982-1988. His works have been performed on concerts and festivals throughout the world, and his music appears on Opus One Records and the Frog Peak Collaborative CD, among others. Miller is an Associate Editor of Electronic Musician magazine, for which he writes about music software and hardware technologies. Since 1998, Miller has also been active as a graphic artist and 3D animator. His animations have been shown at numerous venues throughout the world, most recently the 9th New York Digital Salon, the 2001 Art in Motion screenings, immedia, Sonic Circuits, the Cuban International Festival of Music, and the 2001 Not Still Art screening. His work was also presented at SIGGRAPH 2001 in the Emerging Technologies gallery. Recent exhibits of his 3D still images include the Boston Computer Museum and the Biannual Conference on Art and Technology, as well as publication in Sonic Graphics: Seeing Sound, published by Rizzoli Books. Miller's music and artworks are available at www.casdn.neu.edu/~dmiller.

Pandoremonium

Lissa Meridian

pandoremonium started life as a dance track by San Francisco based duo, Waterjuice. The resulting remix explores deconstruction and reconstruction of the original material, a dissection of the sonic qualities intrinsic in the beat, an implosion and explosion of the space between dance music and more abstract sound material. The mix was performed and recorded in real time at VaporVent studios, San Francisco. The original track, Pandora's Box, was released by Eyefunk Records, and can be found on the album "Hydroponics".

Lissa Meridan (b. 1972) is Director of the Electroacoustic Music Studios at Victoria University of Wellington, where she also teaches instrumental composition, orchestration, counterpoint and acoustics. She is a committee member and webmaster for the Composers Assoc. of NZ and Vice President of the Australasian Computer Music Association.Her works are regularly broadcast on ConcertFM, National Radio and have been performed at concerts and in galleries and festivals in NZ, Australia, USA, Korea, Czechoslovakia, and Bulgaria. She has composed for a wide variety of media, including orchestra, chamber ensembles and electroacoustics. She regularly organises concerts of electroacoustic music and performs live electronic music, both as a sound artist and turntablist (a.k.a. fierce angel). Recent recordings are available on New Zealand Sonic Arts Vol 2, Sonic 2000: New Zealand Electronic Music, and the Atoll release Fanfares for the New Millienium. Technical requirements: CD player

Two Pieces by The Sonic Art Group

Oboe and Additive Gendèr

The *gendèr* is a metallophone found in the Java gamelan orchestra. It is pronounced gendèr (-g) with a "g" as in "get" and with the "er" syllable pronounced "air". The Additive Gendèr is a software instrument - its name referring to the synthesis method used and the characteristic sounds that it produces. The performance of this work features Andrew Ross on oboe and Peter Mcilwain on Additive Gendèr.

Fractured Delay #4

Fractured Delay #4 is from a series of pieces that feature process sound from a variety of sound sources. This piece features a performance on a piano frame that is processed by the fractured delay software. The software uses eight gated delay lines that break up (or fracture) the input sound into segments that are processed by the delay lines. The software can be used for live performance with an eight channel sound system. It can produce complex and varied textures that evolve over time.

The Sonic Art Group is based in the School of Music – Conservatorium at Monash University. Its major role is to function as a research unit directed by Peter Mcilwain – Lecturer in Composition. In addition, it provides opportunities for composition students at the School to gain experience in experimental computer based music. Since starting at the beginning of 2001 it has; performed in a combined concert with Simulation (a group based at QUT) and at the Make It Up Club and has staged the Electronic Antiphony concert. The group's creative work is focused on the composition of sonic textures through improvisation. Much of this work is centred around software developed by the Peter Mcilwain, in conjunction with other members of the group.